

AZ-700: Microsoft Azure Network Engineer Associate

Cheat Sheet

Quick Bytes for you before the exam!

The information provided in the Cheat Sheet is for educational purposes only; it was created in our efforts to help aspirants prepare for the **Exam AZ-700 certification**. Though references have been taken from **Microsoft documentation**, it's not intended as a substitute for the official docs. The document can be reused, reproduced, and printed in any form; ensure that appropriate sources are credited and required permissions are received.

Are you Ready for Microsoft Azure Network Engineer "AZ-700" Certification?

Self-assess yourself with



[Whizlabs FREE TEST](#)



[800+ Hands-on Labs](#)



[Cloud Sandbox Environments](#)

INDEX

Topics Names	Page No
Azure Network Security Services	
Networking Foundation	4
Load Balancing and Content Delivery	4
Hybrid Connectivity	5
Network Security	5
Network Management and Monitoring	5
Implement and manage network security groups	6
Azure Firewall and Azure Firewall Manager	
Azure Firewall	7
Azure Firewall Manager	7
Design & Implement a WAF deployment	
Web Application Firewall [WAF]	9
Azure core networking infrastructure	
Design and implement IP addressing for Azure resources	11
Design and implement name resolution	12
Design and implement VNet connectivity and routing	12
Monitor Networks	13
Azure Network Connectivity Services	
Design, implement, and manage a site-to-site VPN connection	15
Design, implement, and manage a point-to-site VPN connection	15
Azure ExpressRoute	
Design, implement, and manage Azure ExpressRoute	18
Azure Virtual WAN	19

Application Delivery Services	
Azure Load Balancer and Azure Traffic Manager	21
Azure Load Balancer	21
Azure Traffic Manager	21
Azure Application Gateway	23
Azure Front Door	25
Design and implement private access to Azure services	
Design & implement Azure Private Link service and Azure private endpoints	27
Service Endpoints	28
Design and Implement Azure Network Security Services	
1. Create a Network Security Group (NSG)	28
2. Associate an NSG to a Resource	28
3. Create an Application Security Group (ASG)	29
4. Associate an ASG to a Network Interface Card (NIC)	29
5. Create and Configure NSG Rules	29
6. Implement Virtual Network Flow Logs	29
7. Interpret Virtual Network Flow Logs	30
8. Interpret NSG Flow Logs	30
9. Validate NSG Flow Rules	30
10. Verify IP Flow	30
11. Configure an NSG for Remote Server Administration (Including Azure Bastion)	30
12. Implement & Manage Virtual Network Security Using Azure Virtual Network Manager	30

Azure Network Security Services

Azure Networking and Network Security Services	Network Foundations	Azure VNet
		Azure Private Link
		Azure DNS
		Azure Bastion
		Azure NAT Gateway
		Azure Network Watcher
	Connectivity	Azure ExpressRoute
		Azure VPN Gateway
		Azure Virtual WAN
	Application Delivery	Azure Load Balancer
		Azure Application Gateway
		Azure Front Door
	Network Security	Azure Firewall
		Azure DDoS Protection
		Azure Web Application Firewall
		Azure Firewall Manager

Networking Foundation:

Azure's foundational networking services ensure seamless connectivity for your resources. These services include Virtual Network (VNet) for resource isolation, Private Link for private access, Azure DNS for domain management, Azure Bastion for secure remote access, Route Server for dynamic routing, NAT Gateway for outbound internet connectivity, and Traffic Manager for load balancing across geographic locations.

Load Balancing and Content Delivery:

Azure's services for load balancing and content delivery optimize and distribute application workloads. Key offerings include Load Balancer for efficient traffic distribution, Application Gateway for secure application delivery, and Azure Front Door for global content delivery and acceleration.

Hybrid Connectivity:

To enable secure communication between Azure resources and on-premises environments, Azure provides hybrid connectivity services. These include VPN Gateway for secure site-to-site connections, ExpressRoute for private links to Azure data centers, Virtual WAN for simplified networking, and Peering Service for optimized internet connectivity.

Network Security:

Protecting applications and services in Azure is critical. Network security tools include Firewall Manager for centralized policy management, Azure Firewall for network protection, Web Application Firewall (WAF) to safeguard web applications, and DDoS Protection to mitigate distributed denial-of-service attacks.

Network Management and Monitoring:

Azure offers tools to manage and monitor network infrastructure effectively. This includes Network Watcher for diagnostics and insights, Azure Monitor for tracking performance, and Azure Virtual Network Manager for centralized configuration and management.

Network Security Groups (NSGs) act as a firewall at the subnet and NIC level, controlling inbound and outbound traffic based on security rules. They are associated with resources like virtual networks or subnets to enforce policies.

Application Security Groups (ASGs) simplify network security management. Instead of applying NSG rules to individual IPs, you associate ASGs to NICs, creating a logical grouping of resources for better scalability and easier rule management.

NSG Rules are at the heart of traffic control. These rules specify the allowed or denied traffic based on criteria such as IP ranges, ports, or protocols.

Flow Logs (Virtual Network and NSG) provide visibility into traffic patterns. By analyzing these logs, administrators can identify anomalies, validate rules, and troubleshoot issues.

Azure Virtual Network Manager (AVNM) offers centralized control to implement and manage security configurations across multiple virtual networks, ensuring consistency in policy enforcement.

Remote Server Administration using features like Azure Bastion is protected by configuring NSGs to allow secure access while keeping other ports locked down.

Implement and manage network security groups

- **Create a Network Security Group (NSG):** An NSG is a set of security rules used to control network traffic to and from Azure resources. You can create it using Azure Portal, PowerShell, or CLI.
- **Associate an NSG to a Resource:** Once an NSG is created, it can be linked to a subnet or a Network Interface Card (NIC) to apply its rules to that resource's traffic.
- **Create an Application Security Group (ASG):** ASGs allow the logical grouping of resources (e.g., virtual machines) for simplified management of NSG rules, enabling dynamic security policies.
- **Associate an ASG to a NIC:** You can link an ASG to a NIC to apply security rules at a more granular level, targeting specific VM resources.
- **Create and Configure NSG Rules:** NSG rules define traffic flow policies, specifying parameters like source/destination IPs, ports, protocols, and direction (inbound/outbound).
- **Implement Virtual Network Flow Logs:** Flow logs capture network traffic information within virtual networks. They are configured via Azure Network Watcher for monitoring and analysis.
- **Interpret Virtual Network Flow Logs:** Analyze flow logs to understand network traffic patterns, identify bottlenecks, and troubleshoot connectivity issues.
- **Interpret NSG Flow Logs:** These logs provide insights into traffic allowed or denied by NSG rules, helping validate configurations or detect anomalies.
- **Validate NSG Flow Rules:** Use tools like Azure Network Watcher's IP flow verify to test and ensure your NSG rules are configured correctly.
- **Verify IP Flow:** This process checks whether traffic from a source to a destination IP is allowed based on the applied NSG rules.
- **Configure an NSG for Remote Server Administration (e.g., Azure Bastion):** Set up NSG rules to allow secure remote access to servers using Azure Bastion, ensuring ports like RDP or SSH are not directly exposed.
- **Implement and Manage Virtual Network Security with Azure Virtual Network Manager (AVNM):** AVNM provides centralized management for virtual networks, allowing consistent implementation of security and connectivity configurations across resources.

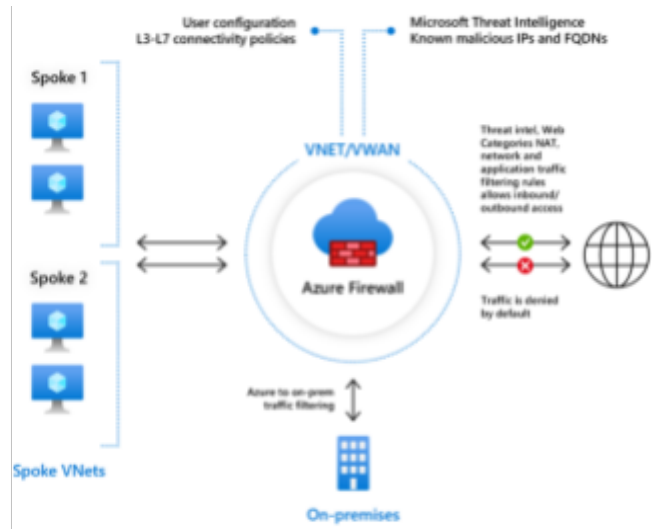
Azure Firewall and Azure Firewall Manager

Azure Firewall

Azure Firewall is a cloud-native, fully stateful network security service designed to protect Azure resources.

It provides Layer 3 to Layer 7 filtering, and threat intelligence-based filtering, and supports features like NAT (Network Address Translation), application rules, and network rules.

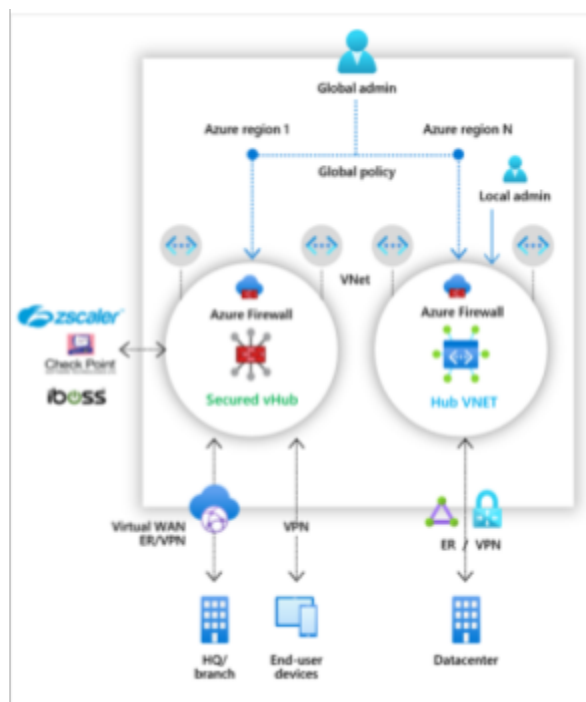
Azure Firewall is scalable and integrates seamlessly with other Azure services to ensure robust security for both inbound and outbound traffic.



Azure Firewall Manager

Azure Firewall Manager is a centralized security management service that simplifies the deployment and configuration of Azure Firewalls across multiple subscriptions and regions.

It allows you to create and enforce hierarchical policies, manage secured virtual hubs, and integrate with third-party security solutions. This tool streamlines the management of network security policies, ensuring consistency and compliance across your Azure environment.



Source: Microsoft Documentation

1. **Map Requirements to Features and Capabilities of Azure Firewall:** Azure Firewall provides essential capabilities, such as threat intelligence, application and network filtering, NAT rules, and FQDN filtering. Requirements like scalability, high availability, or centralized

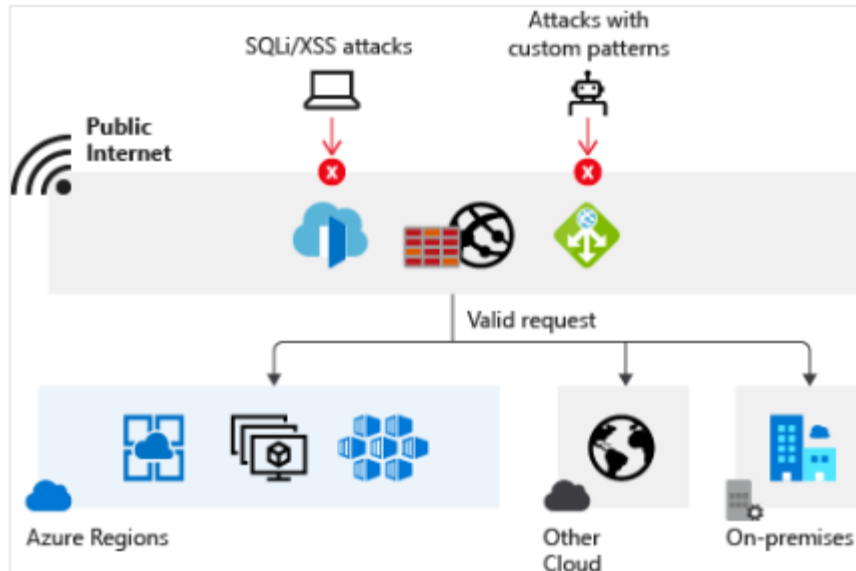
policy management are matched with Azure Firewall features to meet specific business needs.

2. **Select an Appropriate Azure Firewall SKU:** Azure Firewall offers two SKUs—Standard and Premium. The Standard SKU is suitable for general security use cases, while the Premium SKU includes advanced features like TLS inspection, IDPS (Intrusion Detection and Prevention System), and URL filtering, making it ideal for complex security needs.
3. **Design an Azure Firewall Deployment:** Designing includes selecting deployment models (hub-spoke, secured virtual hub, etc.), determining network architecture, specifying subnets for the firewall, and integrating other Azure services such as Azure Monitor or Threat Intelligence.
4. **Create and Implement an Azure Firewall Deployment:** Deploy the Azure Firewall using Azure Portal, PowerShell, or CLI, configure the virtual network and subnets, and integrate the firewall with existing resources for traffic filtering.
5. **Configure Azure Firewall Rules:** Create and manage rules to define traffic flow policies. These include application rules for managing app traffic, network rules for protocol-based filtering, and NAT rules to translate traffic addresses.
6. **Create and Implement Azure Firewall Manager Policies:** Azure Firewall Manager enables centralized policy creation and enforcement across multiple firewalls. This simplifies configuration by using hierarchies and templates for consistent management.
7. **Create a Secure Hub by Deploying Azure Firewall Inside an Azure Virtual WAN Hub:** Deploying an Azure Firewall within an Azure Virtual WAN hub enhances security by centralizing traffic inspection and applying security policies across spoke networks, ensuring secure communication between hybrid and cloud environments.

Design & Implement a WAF deployment

Web Application Firewall [WAF]

It is a security solution designed to protect web applications by filtering and monitoring HTTP traffic between the application and the internet. It defends against common threats such as SQL injection, cross-site scripting (XSS), and other application-layer attacks.



Source: Microsoft Documentation

WAF Deployment can be implemented in various ways:

1. **Network-Based WAF:** Deployed at the network level, offering minimal latency but requiring physical hardware.
2. **Host-Based WAF:** Integrated into the application software, providing flexibility and customization.
3. **Cloud-Based WAF:** Hosted in the cloud, offering scalability and ease of deployment without the need for on-premises infrastructure.

By deploying a WAF, organizations can create a protective shield for their web applications, ensuring secure and uninterrupted access for users.

Map Requirements to Features and Capabilities of WAF: Match your application security needs (e.g., protection against OWASP Top 10 threats, geo-filtering, or bot mitigation) with WAF features like managed rule sets, custom rule configurations, and integration with Azure services.

Design a WAF Deployment: Plan the deployment by selecting a WAF type—Azure Front Door or Application Gateway—based on traffic distribution needs and application architecture. Ensure scalability, high availability, and integration with monitoring tools.

Configure Detection or Prevention Mode: In detection mode, WAF only logs potential threats without blocking traffic. Prevention mode actively blocks malicious traffic based on predefined and custom rules, ensuring real-time protection.

Configure Rule Sets for WAF on Azure Front Door: Apply managed rule sets for common vulnerabilities like XSS or SQL injection. Customize rules to meet specific application requirements while enabling global content delivery.

Configure Rule Sets for WAF on Application Gateway: Customize rules for inbound traffic inspection at the regional gateway level. Define conditions such as IP restrictions, request size limits, and protocol validation to secure applications.

Implement a WAF Policy: Create a WAF policy defining the security rules and settings to be applied. This policy centralizes management and enforces consistent configurations across all WAF deployments.

Associate a WAF Policy: Link the WAF policy to specific Azure Front Door or Application Gateway instances to apply the rules and settings, ensuring uniform protection across applications.

Azure core networking infrastructure

Design and implement IP addressing for Azure resources

Plan and Implement Network Segmentation and Address Spaces: Design network segmentation to isolate and secure resources. Allocate IP address ranges and segments within virtual networks (VNETs) while avoiding overlapping ranges for seamless communication.

Create a Virtual Network (VNet): A VNet is the foundation of Azure networking. It provides an isolated environment for deploying and connecting resources securely, supporting subnets, firewalls, and more.

Plan and Configure Subnetting for Services: Subnets are segments within a VNet for organizing resources. Configure them to support specific services like VNet gateways (for connectivity), private endpoints (for private access), service endpoints, firewalls, application gateways, and Azure Bastion for secure access.

Plan and Configure Subnet Delegation: Assign specific subnets to managed services (e.g., Azure Kubernetes Service) to simplify provisioning and integration.

Plan and Configure Shared or Dedicated Subnets: Shared subnets host multiple resources, while dedicated subnets are allocated to specific services for enhanced isolation and security.

Create a Prefix for Public IP Addresses: Reserve a range of public IP addresses (prefix) for resource allocation, ensuring availability and simplifying address management.

Choose When to Use a Public IP Address Prefix: Use public IP address prefixes when managing multiple resources that require consistent and scalable public-facing access.

Plan and Implement a Custom Public IP Address Prefix (Bring Your Own IP): Bring your own IP addresses to Azure to maintain existing IP relationships and ensure smooth migration to cloud environments.

Create a Public IP Address: Assign a globally unique public IP to resources for internet-facing connectivity, used in scenarios like hosting web applications.

Associate Public IP Addresses to Resources: Bind public IP addresses to specific resources (e.g., virtual machines or load balancers) to enable external access.

Upgrade IP Address SKU: Upgrade from basic to standard SKU to benefit from advanced features like zone redundancy, availability across regions, and better reliability.

Design and implement name resolution

Design Name Resolution Inside a VNet: Plan how DNS resolves names within a virtual network. This involves selecting between Azure's default DNS, custom DNS servers, or private DNS zones for managing name resolution.

Configure DNS Settings for a VNet: Specify custom DNS server IPs for the VNet, or use Azure-provided DNS for automatic name resolution. These settings apply to resources deployed in the network.

Design Public DNS Zones: Public DNS zones handle internet-facing domain names. Design these zones to map domain names to public IP addresses of Azure resources, ensuring global accessibility.

Design Private DNS Zones: Private DNS zones manage name resolution within the internal network, providing secure DNS mappings for Azure resources that are not exposed to the internet.

Configure Public and Private DNS Zones: Set up public DNS zones for externally accessible domains and private DNS zones for internal resolution. Configure records like A, CNAME, and MX appropriately for each zone.

Link a Private DNS Zone to a VNet: Connect a private DNS zone to a VNet to enable automatic name resolution for resources within that network without requiring custom DNS servers.

Design and Implement Azure DNS Private Resolver: Azure DNS Private Resolver provides name resolution between on-premises and Azure resources. Design it to integrate with private DNS zones and configure endpoints for secure and seamless name resolution.

Design and implement VNet connectivity and routing

Design Service Chaining, Including Gateway Transit: Service chaining involves directing traffic through specific network appliances (e.g., firewalls or monitoring tools) within a virtual network. Gateway transit enables connected VNets to share the same VPN or ExpressRoute gateway for optimized connectivity.

Implement VNet Peering: VNet peering connects two virtual networks, allowing seamless communication between resources without using a public gateway. It ensures low latency and high bandwidth across peered networks.

Implement and Manage Virtual Network Connectivity Using Azure Virtual Network Manager:

Azure Virtual Network Manager centralizes management of virtual networks, enabling seamless implementation of network configurations and connectivity across multiple VNets and regions.

Design and Implement User-Defined Routes (UDRs): UDRs allow customization of traffic flow by specifying route paths for packets, enabling control over how traffic reaches its destination in a VNet or between VNets.

Associate a Route Table with a Subnet: Route tables define traffic routing rules. By associating a route table with a subnet, you apply these rules to direct network traffic within that segment.

Configure Forced Tunneling: Forced tunneling ensures all outbound traffic from an Azure resource is routed through an on-premises network, enhancing security and compliance by using a VPN gateway.

Diagnose and Resolve Routing Issues: Use tools like Azure Network Watcher to identify and troubleshoot routing problems, ensuring network traffic follows the intended paths and avoiding misconfigurations.

Design and Implement Azure Route Server: Azure Route Server simplifies dynamic routing between on-premises networks and Azure VNets by automatically exchanging routes with network appliances.

Identify Appropriate Use Cases for a NAT Gateway: NAT gateways are ideal for providing outbound internet access for resources in private subnets without exposing their private IP addresses to the internet.

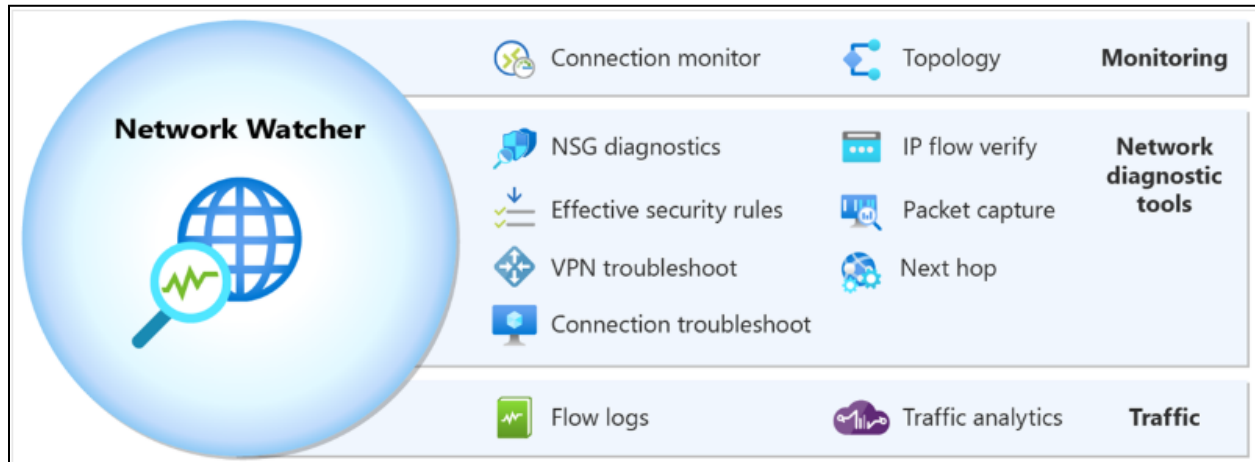
Implement a NAT Gateway: Deploy a NAT gateway in your virtual network and associate it with one or more subnets to enable secure outbound connectivity for your Azure resources.

Monitor Networks

Azure Network Watcher is a comprehensive networking tool designed to monitor, diagnose, and optimize the health of Azure virtual networks. It offers features like connection monitoring, topology visualization, packet capture, and IP flow verification, helping administrators troubleshoot connectivity issues. Network Watcher also supports network security group (NSG) flow logs for analyzing traffic patterns and identifying potential risks. By providing real-time insights and diagnostics, it ensures efficient management and high performance of Azure network resources.

Network Watcher consists of three major sets of tools and capabilities:

- Monitoring
- Network diagnostic tools
- Traffic



Source: Microsoft Documentation

Configure Monitoring, Network Diagnostics, and Logs in Azure Network Watcher: Azure Network Watcher provides tools to monitor and diagnose networking issues. Configure logging features like flow logs, packet capture, and diagnostic tools to gain insights into traffic and troubleshoot problems effectively.

Monitor and Troubleshoot Network Health Using Azure Network Watcher: Use features such as Connection Monitor, IP Flow Verify, and Topology Viewer to detect and resolve network issues. These tools provide a real-time view of your network's health and performance.

Monitor and Troubleshoot Networks Using Azure Monitor Network Insights: Azure Monitor Network Insights offers a centralized view of network resources and health. Utilize metrics, diagnostic logs, and alerts to monitor network performance and address potential bottlenecks.

Activate and Monitor Distributed Denial-of-Service (DDoS) Protection: Enable Azure DDoS Protection to safeguard applications from large-scale attacks. Monitor DDoS activity using Azure Monitor and take actions based on telemetry data and insights.

Evaluate Network Security Recommendations Identified by Microsoft Defender for Cloud Secure Score: Assess your network's security posture using the Secure Score feature, which provides actionable recommendations to address vulnerabilities and improve compliance.

Evaluate Network Security Recommendations Identified by Microsoft Defender for Cloud Attack Path Analysis: Use Attack Path Analysis to identify potential attack vectors in your network. Follow recommendations to strengthen security and reduce risks.

Identify Network Resources Using Microsoft Defender for Cloud Security Explorer: Security Explorer helps locate and assess the security state of network resources, providing insights into misconfigurations or vulnerabilities for proactive resolution.

Azure Network Connectivity Services

Design, implement, and manage a site-to-site VPN connection

Design a Site-to-Site VPN Connection, Including High Availability: Plan the connection by designing redundant VPN gateways and paths to ensure high availability. Use Azure's Active-Active VPN gateway configuration to avoid single points of failure.

Select an Appropriate VNet Gateway SKU for Site-to-Site VPN Requirements: Choose between Azure VPN Gateway SKUs (Basic, Standard, High Performance, or Ultra) based on throughput, concurrent connections, and availability requirements.

Implement a Site-to-Site VPN Connection: Establish a secure connection between on-premises and Azure VNets by configuring the VPN gateway, local network gateway, and connection settings on both ends.

Identify When to Use a Policy-Based VPN Versus a Route-Based VPN Connection: Use a Policy-Based VPN for static routing scenarios with fewer connections, while a Route-Based VPN is better for dynamic routing, multiple tunnels, and flexibility.

Create and Configure a Local Network Gateway: Define on-premises network details (such as public IP and address space) in Azure to represent your local network for VPN connections.

Create and Configure an IPsec/Internet Key Exchange (IKE) Policy: Customize encryption and authentication algorithms for VPN tunnels by creating an IKE policy, ensuring secure communication.

Create and Configure a Virtual Network Gateway: Deploy a virtual network gateway in a subnet, configure its SKU, and associate it with your virtual network for routing and connectivity.

Diagnose and Resolve Virtual Network Gateway Connectivity Issues: Use tools like Azure Network Watcher to troubleshoot common issues, such as misconfigured routes, IP address conflicts, or connectivity errors.

Implement Azure Extended Network: Extend your on-premises network to Azure by deploying the Azure Extended Network feature, which allows workloads to seamlessly span both environments.

Design, implement, and manage a point-to-site VPN connection

Select an Appropriate Virtual Network Gateway SKU for Point-to-Site VPN Requirements:

Choose a gateway SKU like Basic, Standard, or High Performance based on features such as concurrent connection limits, supported protocols, and throughput for your point-to-site VPN.

Select and Configure a Tunnel Type: Decide between SSTP (Secure Socket Tunneling Protocol), OpenVPN, or IKEv2 based on compatibility, security needs, and client support. Configure the tunnel type during gateway setup.

Select an Appropriate Authentication Method: Choose between certificate-based authentication, RADIUS authentication, or Microsoft Entra ID for VPN users, depending on your organization's security policies and infrastructure.

Configure RADIUS Authentication: Set up a RADIUS server for centralized authentication. Configure the VPN gateway to communicate with this server, allowing user access to be validated.

Configure Authentication by Using Microsoft Entra ID: Integrate Microsoft Entra ID (formerly Azure Active Directory) with your VPN gateway to enable modern, cloud-based authentication methods such as multi-factor authentication (MFA).

Implement a VPN Client Configuration File: Generate and distribute a configuration file for VPN clients. This file includes server details, authentication settings, and certificate information for client connectivity.

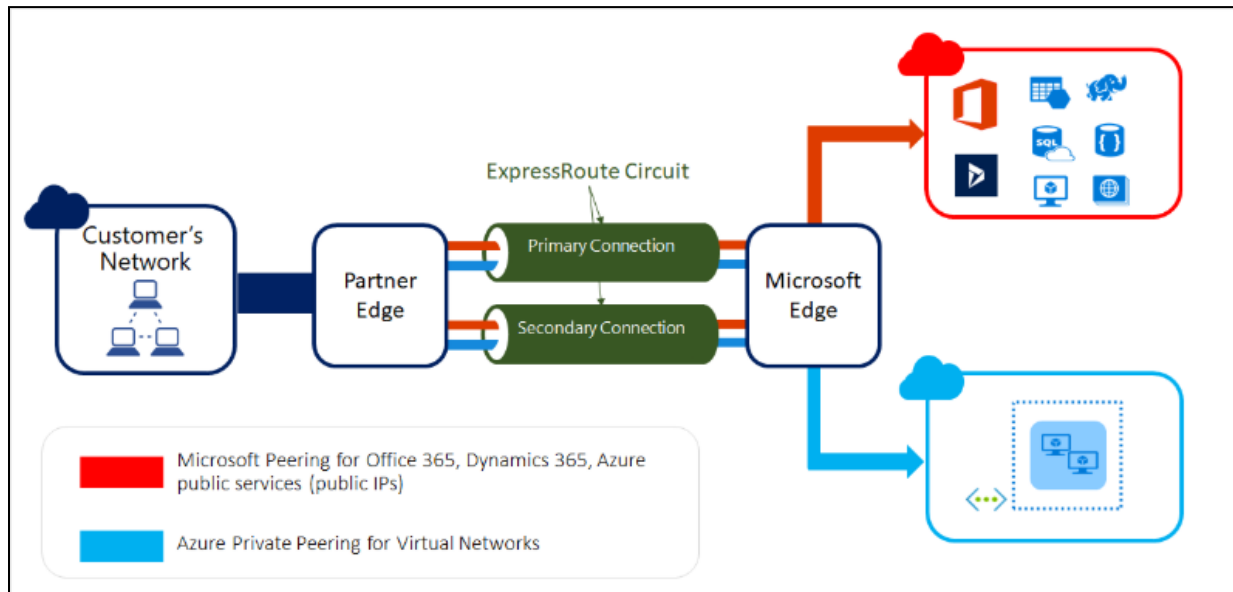
Diagnose and Resolve Client-Side and Authentication Issues: Use tools like logs, connection diagnostics, and Network Watcher to identify and troubleshoot errors related to client configuration, connectivity, or authentication.

Specify Azure Requirements for Always On VPN: Ensure resources like VPN gateways, appropriate tunnel types, and security policies are configured to support Always On VPN for uninterrupted connectivity.

Specify Azure Requirements for Azure Network Adapter: Deploy and configure the Azure Network Adapter to establish secure point-to-site connections from Windows Server clients to Azure resources.

Azure ExpressRoute

Azure ExpressRoute is a dedicated private connection service that establishes secure and high-performance connectivity between your on-premises infrastructure and Microsoft Azure. Unlike typical internet connections, ExpressRoute offers higher reliability, lower latencies, and enhanced security by creating a direct link to Azure's data centers.



Source: Microsoft Documentation

Key Features:

- **High Bandwidth and Scalability:** Provides support for data-intensive workloads with high throughput requirements.
- **Enhanced Security:** Keeps traffic off the public internet, ensuring secure and private communication.
- **Global Connectivity with ExpressRoute Global Reach:** Allows interconnection between on-premises sites through Azure's global network.
- **Support for Redundancy and Disaster Recovery:** Offers redundancy through Active-Active connections and failover capabilities to ensure business continuity.

Use Cases:

- Running enterprise-grade applications in Azure with consistent performance.
- Migrating large amounts of data securely and reliably to Azure.
- Connecting hybrid environments for seamless integration between on-premises resources and cloud services.

Design, implement, and manage Azure ExpressRoute

Select an ExpressRoute Connectivity Model: Choose between **Point-to-Point Ethernet**, **Any-to-Any (IPVPN)**, or **Direct Connection** models based on your organization's existing infrastructure and connectivity requirements.

Select an Appropriate ExpressRoute SKU and Tier: Pick from **Local**, **Standard**, or **Premium** SKUs, and choose appropriate tiers based on bandwidth needs, global connectivity requirements, and features like multiple regions access.

Design and Implement ExpressRoute to Meet Requirements: Implement ExpressRoute circuits with redundancy (using Active-Active gateways) and cross-region connectivity. Factor disaster recovery into the design using paired regions and failover configurations.

Design and Implement ExpressRoute Options: Use **Global Reach** for private network interconnectivity across locations, **FastPath** to optimize data path performance, and **ExpressRoute Direct** for dedicated access with higher bandwidth.

Choose Between Azure Private Peering, Microsoft Peering, or Both: Decide based on the traffic type. Private peering is used for VNet access, while Microsoft peering is for SaaS services (e.g., Microsoft 365). Both can be enabled for broader requirements.

Configure Azure Private Peering: Set up private peering for secure access to Azure resources within VNets. Define IP ranges and configure access routes accordingly.

Configure Microsoft Peering: Set up Microsoft Peering for accessing Microsoft SaaS services by configuring the required AS numbers and IP addresses.

Create and Configure an ExpressRoute Gateway: Deploy the gateway in a dedicated subnet of your VNet. Configure the gateway SKU and connect it to the ExpressRoute circuit.

Connect a Virtual Network to an ExpressRoute Circuit: Establish the connection between the VNet and the ExpressRoute circuit, ensuring route propagation is enabled for seamless communication.

Recommend a Route Advertisement Configuration: Use optimized route advertisement to avoid conflicts, such as splitting address spaces to ensure efficient traffic routing.

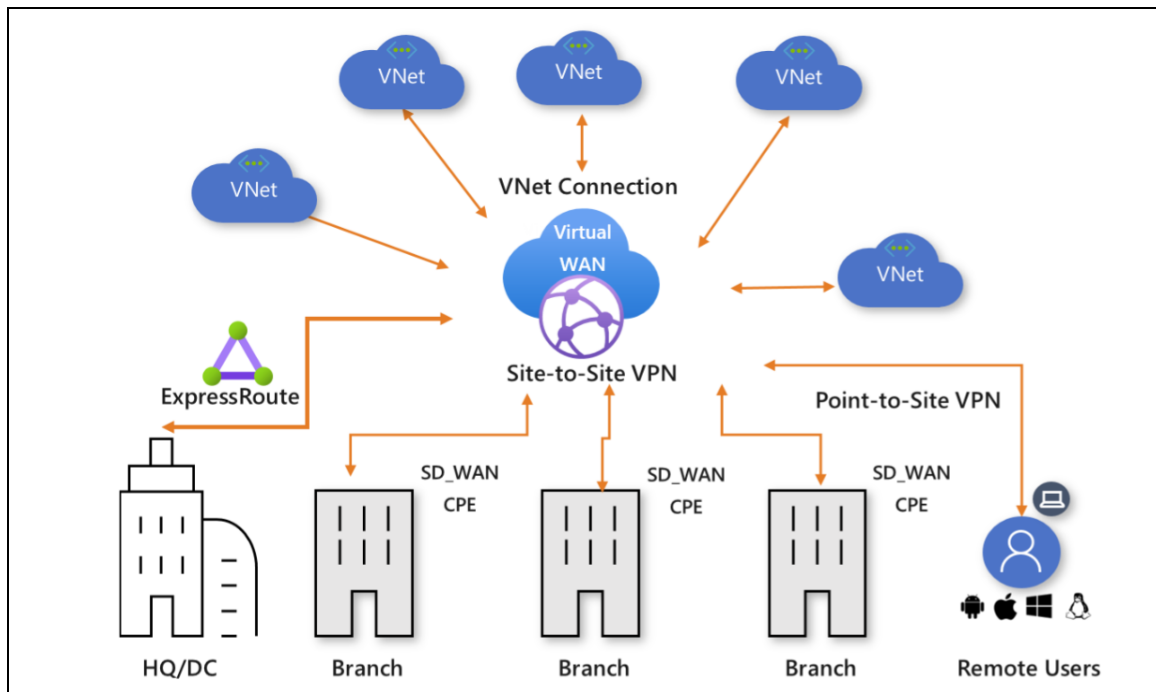
Configure Encryption Over ExpressRoute: Secure traffic by implementing IPsec tunnels for encryption over ExpressRoute private connections when necessary.

Implement Bidirectional Forwarding Detection (BFD): Use BFD to enhance the reliability of the ExpressRoute circuit by detecting failures in forwarding paths quickly.

Diagnose and Resolve ExpressRoute Connection Issues: Use tools like Azure Network Watcher, diagnostics logs, and troubleshooting guides to identify and fix connectivity problems.

Azure Virtual WAN

Azure Virtual WAN is a networking service that simplifies and enhances connectivity across on-premises, cloud, and branch environments. Its architecture is based on a **hub-and-spoke model**, where the hub acts as a central point for connectivity and routing.



Source: Microsoft Documentation

Here's a brief overview:

1. **Hub:** The Azure Virtual WAN hub is a managed resource that connects various spokes, such as virtual networks (VNets), branch sites, and remote users. It supports features like VPN, ExpressRoute, and Azure Firewall for secure and efficient communication.
2. **Spokes:** These are virtual networks or branch sites connected to the hub. Spokes can communicate with each other through the hub, enabling transitive connectivity.
3. **Global Transit Network:** Azure Virtual WAN enables a global transit network, allowing seamless connectivity between hubs in different Azure regions. This ensures low-latency communication and optimized routing.
4. **Security and Routing:** The architecture integrates Azure Firewall and routing policies to enhance security and simplify traffic management.
5. **Scalability:** The hub-and-spoke design supports large-scale deployments, making it suitable for enterprises with complex networking needs.

Select a Virtual WAN SKU: Azure Virtual WAN offers two SKUs: Basic and Standard. The Basic SKU supports site-to-site VPN only, while the Standard SKU includes additional features like ExpressRoute, point-to-site VPN, and hub-to-hub connectivity. Choose the SKU based on your connectivity and scalability needs.

Design a Virtual WAN Architecture: Use a hub-and-spoke model where the Virtual WAN hub acts as the central point for connectivity. Integrate services like Azure Firewall, VPN gateways, and ExpressRoute for secure and efficient communication. Ensure the architecture supports global transit and scales across regions.

Create a Hub in Virtual WAN: A Virtual WAN hub is a Microsoft-managed virtual network. Create it in the Azure portal by specifying the region, address space, and required gateways (e.g., VPN, ExpressRoute). The hub serves as the core of your network.

Choose an Appropriate Scale Unit for Each Gateway Type: Scale units determine the throughput of gateways. For example, site-to-site VPN gateways can scale from 500 Mbps to higher capacities. Select scale units based on the expected traffic and redundancy requirements.

Deploy a Gateway into a Virtual WAN Hub: Deploy gateways (e.g., VPN, ExpressRoute) into the hub to enable connectivity. Configure settings like IP ranges and routing preferences during deployment. Gateways facilitate communication between on-premises, cloud, and branch locations.

Configure Virtual Hub Routing: Use the hub router to manage traffic between connected resources. Create route tables to define traffic paths and associate them with connections. Configure routing policies for internet-bound and private traffic to optimize security and performance.

Integrate a Virtual WAN Hub with a Third-Party NVA: Deploy Network Virtual Appliances (NVAs) like SD-WAN or firewalls into the hub for advanced connectivity and security. NVAs integrate seamlessly with the hub's routing system, enabling traffic inspection and policy enforcement.

Application Delivery Services

Azure Load Balancer and Azure Traffic Manager

Azure Load Balancer:

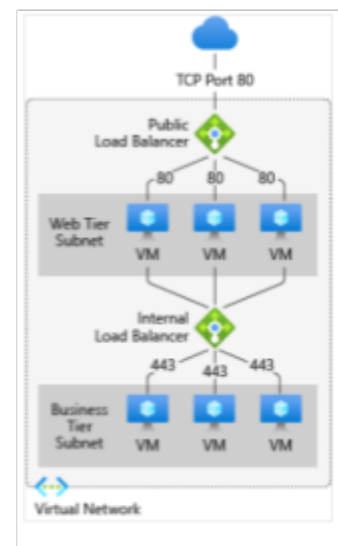
A Layer 4 load balancer designed to distribute incoming traffic across multiple backend resources like Virtual Machines (VMs). Offers **high availability**, **low latency**, and **automatic scaling** for applications. This will support both **Public Load Balancers** (for internet-facing traffic) and **Internal Load Balancers** (for internal network traffic)

Available in **Basic** and **Standard** SKUs—Standard provides advanced features such as zone redundancy, diagnostics, and higher scalability.

You can create and configure **load balancing rules**, **inbound NAT rules**, and **outbound SNAT rules** for specific traffic management.

Use Cases:

- ◆ Distributing web application traffic.
- ◆ Scaling backend services.
- ◆ Improving fault tolerance for applications.



Source: Microsoft Documentation

Azure Traffic Manager:

A **DNS-based traffic routing solution** that directs user requests to the most appropriate endpoint. Enables **global load balancing** and ensures high availability for distributed applications.

Map Requirements to Features and Capabilities of Azure Load Balancer: Azure Load Balancer operates at Layer 4 (TCP/UDP) and distributes incoming traffic across backend resources. It supports high availability, low latency, and scalability. Features include health probes, session persistence, and support for both inbound and outbound traffic.

Identify Appropriate Use Cases for Azure Load Balancer: Use Azure Load Balancer for scenarios requiring high availability and load distribution, such as balancing traffic across virtual machines, enabling outbound internet connectivity, or supporting hybrid cloud environments.

Supports multiple routing methods:

Priority: Routes to the primary endpoint unless unavailable.

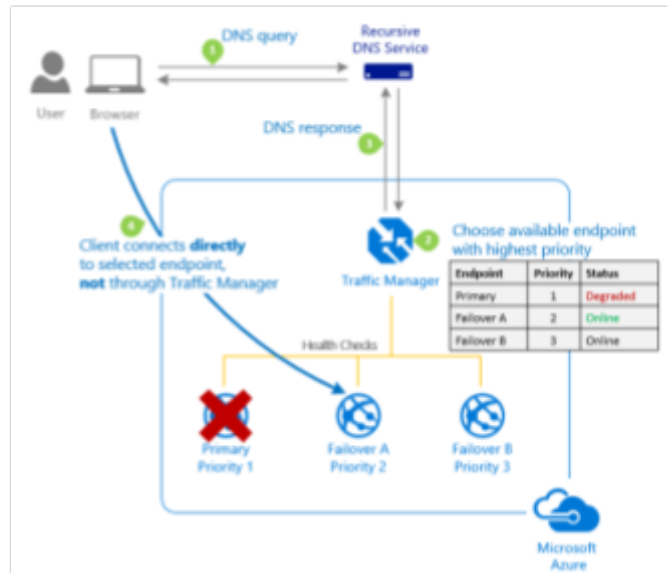
Performance: Sends users to the endpoint with the lowest latency.

Weighted: Distributes traffic based on assigned weights.

Geographic: Routes based on user location.

Use Cases:

- Directing users to regional data centers for better performance.
- Implementing disaster recovery by routing to backup regions during outages.



Choose an Azure Load Balancer SKU and Tier: Azure Load Balancer offers Basic, Standard, and Gateway SKUs. The Basic SKU is suitable for small-scale applications, while the Standard SKU supports advanced features like zone redundancy and diagnostics. Gateway SKU is designed for scenarios involving Network Virtual Appliances (NVAs).

Choose Between Public and Internal Load Balancers: Public Load Balancers handle internet-facing traffic, while Internal Load Balancers manage traffic within a virtual network. Choose based on whether the traffic is external or internal.

Choose Between Regional and Global Load Balancers: Regional Load Balancers distribute traffic within a single Azure region, while Global Load Balancers enable cross-region traffic distribution for geo-redundancy and failover.

Create and Configure an Azure Load Balancer: Set up a Load Balancer by defining frontend IP configurations, backend pools, health probes, and load-balancing rules. Use the Azure portal, CLI, or PowerShell for configuration.

Implement Azure Traffic Manager: Azure Traffic Manager is a DNS-based traffic load balancer that routes traffic to endpoints based on routing methods like Priority, Weighted, Performance, and Geographic. It ensures high availability and responsiveness for global applications.

Implement a Gateway Load Balancer: A Gateway Load Balancer integrates seamlessly with NVAs for scenarios like firewalls or intrusion detection. It ensures traffic symmetry and simplifies the deployment of NVAs.

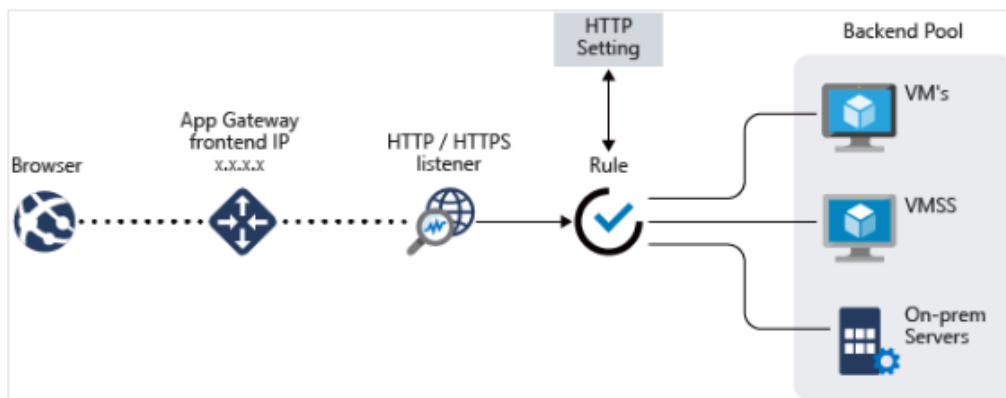
Implement a Load Balancing Rule: Load balancing rules define how traffic is distributed across backend resources. Configure rules by specifying frontend IP, backend pool, protocol, and port settings.

Create and Configure Inbound NAT Rules: Inbound NAT rules forward traffic from a Load Balancer frontend to specific backend resources. They are used for port forwarding to enable direct access to virtual machines.

Create and Configure Explicit Outbound Rules, Including SNAT: Explicit outbound rules define how backend resources connect to the internet using Source Network Address Translation (SNAT). Configure these rules to control outbound traffic and optimize IP usage.

Azure Application Gateway

It's A Layer 7 load balancer that manages web traffic to your applications.



→ Key Features:

- ◆ **URL-based routing:** Directs traffic based on URL paths for hosting multiple web applications.
- ◆ **Web Application Firewall (WAF):** Protects against common web vulnerabilities like SQL injection.
- ◆ **SSL Termination:** Offloads SSL decryption for backend performance.
- ◆ **Autoscaling:** Handles traffic spikes seamlessly.
- ◆ **Header Rewrite:** Modifies HTTP request and response headers.

→ **Use Cases:**

- ◆ Distributing traffic to backend services in web applications.
- ◆ Enhancing security with WAF.
- ◆ Managing traffic within a virtual network (internal gateway).

Map Requirements to Features and Capabilities of Azure Application Gateway: Azure Application Gateway provides Layer 7 load balancing, URL-based routing, SSL termination, Web Application Firewall (WAF), and autoscaling. These features align with requirements for secure, scalable, and intelligent traffic management.

Identify Appropriate Use Cases for Azure Application Gateway: Ideal for hosting multiple web applications, securing applications with WAF, enabling SSL offloading, and managing traffic for stateful or stateless applications.

Choose Between Manual and Autoscale:

Manual Scaling: Fixed instance count, suitable for predictable workloads.

Autoscaling: Dynamically adjusts capacity based on traffic, ideal for fluctuating workloads.

Create a Back-End Pool: Define backend targets (e.g., VMs, IP addresses, or App Services) to distribute traffic. Configure these pools in the Azure portal under the Application Gateway settings.

Configure Health Probes: Set up health probes to monitor backend server health. Use custom probes for specific requirements or default probes for basic monitoring.

Configure Listeners: Listeners check for incoming requests on specified ports and protocols. Configure basic or multi-site listeners based on your application needs.

Configure Routing Rules: Define rules to route traffic to backend pools. Use basic rules for single backend pools or path-based rules for URL-specific routing.

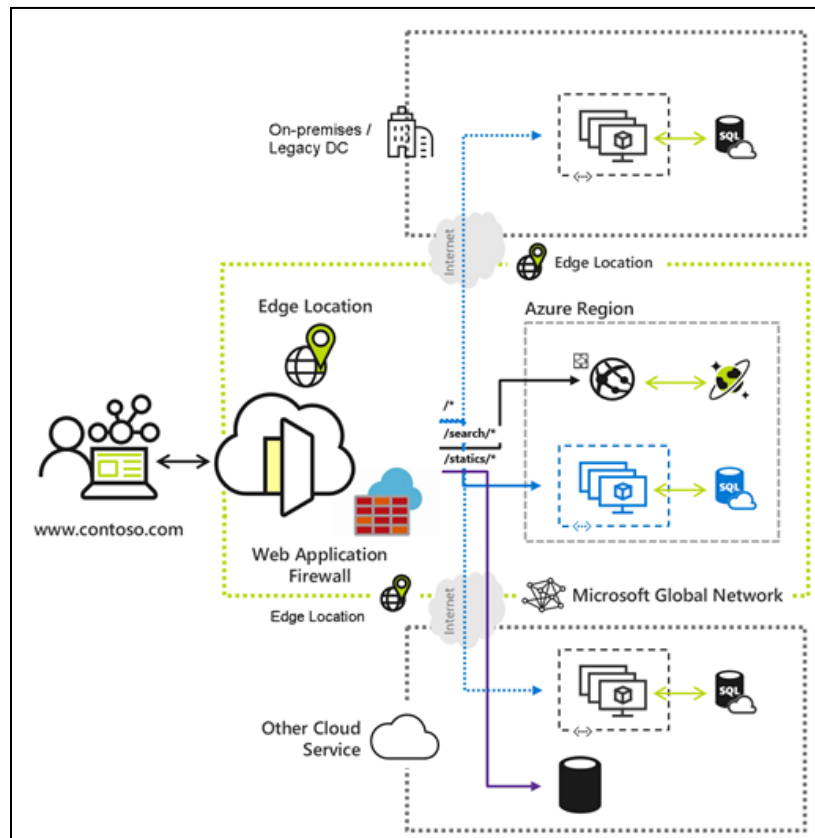
Configure HTTP Settings: Specify backend communication settings, including protocol, port, and session affinity. Enable connection draining for seamless updates.

Configure Transport Layer Security (TLS): Set up TLS termination for secure traffic handling. Use predefined or custom TLS policies to meet compliance and security standards.

Configure Rewrite Sets: Modify HTTP headers or URLs for requests and responses. Use rewrite sets to add, remove, or update headers and rewrite URLs based on conditions.

Azure Front Door

A global, scalable service for fast delivery and secure access to web applications.



→ Key Features:

- ◆ **Content delivery network (CDN):** Caches and delivers content globally for lower latency.
- ◆ **Global load balancing:** Routes user traffic to the nearest or healthiest backend.
- ◆ **HTTPS for Custom Domains:** Simplifies SSL management for custom domains.
- ◆ **Path-based Routing:** Routes traffic based on URL paths.
- ◆ **Session Affinity:** Keeps users connected to the same backend server.

→ Use Cases:

- ◆ Improving application performance by reducing latency for global users.
- ◆ Implementing disaster recovery through multi-region failover.
- ◆ Accelerating content delivery with caching.

Map Requirements to Features and Capabilities of Azure Front Door: Azure Front Door provides global load balancing, content delivery network (CDN) capabilities, SSL termination, caching, and traffic acceleration. It supports routing based on latency, priority, and weight, ensuring optimal performance and security for web applications.

Identify Appropriate Use Cases for Azure Front Door: Ideal for scenarios requiring global load balancing, fast content delivery, disaster recovery, and secure access to web applications. It's also suitable for applications with high traffic or distributed user bases.

Choose an Appropriate Tier: Azure Front Door offers Standard and Premium tiers. The Standard tier focuses on content delivery and basic security, while the Premium tier includes advanced features like Web Application Firewall (WAF), Private Link, and bot protection. Choose based on your security and performance needs.

Configure an Azure Front Door (Routing, Origins, and Endpoints): Set up routing rules to direct traffic to backend origins. Define origins (e.g., Azure App Services, Storage) and endpoints (e.g., custom domains) to manage traffic flow effectively.

Configure SSL Termination and End-to-End SSL Encryption: Enable SSL termination at the edge for secure traffic handling. Use end-to-end SSL encryption to ensure data security from the client to the origin server.

Configure Caching: Enable caching for static content like images and scripts to reduce latency and server load. Configure caching rules to optimize performance and avoid caching sensitive data.

Configure Traffic Acceleration: Use Azure Front Door's traffic acceleration features, such as split TCP and Anycast, to reduce latency and improve performance for global users.

Implement Rules, URL Rewrite, and URL Redirect: Use the rules engine to define custom routing behaviors. Implement URL rewrite to modify request paths and URL redirect to guide users to new URLs or enforce HTTPS.

Secure an Origin Using Azure Private Link: In the Premium tier, use Azure Private Link to connect the Front Door to private origins securely. This eliminates public exposure and ensures traffic flows through the Microsoft backbone network.

Design and implement private access to Azure services

Designing and implementing private access to Azure services involves creating secure connections between your virtual networks and Azure resources without exposing them to the public internet. This can be achieved using **Azure Private Link** and **virtual network service endpoints**.

- **Azure Private Link:** This allows you to connect to Azure services like Azure Storage or Azure SQL Database through a private endpoint within your virtual network. It ensures that traffic remains within the Azure backbone network, enhancing security and reducing exposure to threats.
- **Service Endpoints:** These extend your virtual network's private address space to Azure services, enabling secure access while maintaining optimal performance.

Design and implement Azure Private Link service and Azure private endpoints

Plan Private Endpoints: Planning involves identifying which Azure resources require secure access via private endpoints. Assess network architecture, ensure subnet availability within your virtual network, and evaluate DNS integration requirements for seamless connectivity.

Create Private Endpoints: To create private endpoints, navigate to the desired Azure resource, configure a private endpoint, and assign it to a subnet within your virtual network. This ensures secure traffic routing without public internet exposure.

Configure Access to Private Endpoints: Fine-tune access by managing network security groups (NSGs) and firewall rules. Implement role-based access control (RBAC) to define who can access the resources linked to private endpoints.

Create a Private Link Service: A Private Link service allows you to offer your service privately to consumers within their virtual networks. It involves configuring a load balancer and enabling private connectivity through the Azure backbone network.

Integrate Private Link and Private Endpoint with DNS: For seamless connectivity, set up private DNS zones and link them to your virtual networks. Ensure proper resolution of private endpoint DNS names to their private IP addresses.

Integrate a Private Link Service with On-Premises Clients: This involves extending private connectivity to on-premises networks using VPN gateways or ExpressRoute. Configure DNS forwarding and hybrid network solutions for consistent private access.

Service Endpoints

Choose When to Use a Service Endpoint: Service endpoints are best used when you need secure, high-performance access to Azure services over the Azure backbone network. They are particularly suitable for scenarios where you want to restrict access to Azure services from specific virtual networks without requiring public IP addresses.

Create Service Endpoints: To create service endpoints, enable them on the subnet associated with your virtual network. This can be done through the Azure portal, PowerShell, or CLI. Once enabled, traffic from that subnet to the selected Azure service will use the private Azure network.

Configure Service Endpoint Policies: Service endpoint policies allow you to control access to specific Azure service instances. Define rules within the policy to restrict or permit traffic from your virtual network to particular resources (e.g., specific storage accounts), providing granular access control.

Configure Access to Service Endpoints: Ensure access by aligning network security groups (NSGs) and firewalls with your service endpoint configuration. Set permissions using IAM to allow only authorized users and applications to interact with the connected resources.

Design and Implement Azure Network Security Services

1. Create a Network Security Group (NSG)

A Network Security Group (NSG) acts as a virtual firewall to control inbound and outbound traffic at the subnet or NIC level in Azure. To create one:

- Go to the Azure portal and search for **Network Security Groups**.
- Select **Create** and provide the required details like NSG name, resource group, and region.
- Once created, it will initially have default rules to allow essential traffic and block everything else.

2. Associate an NSG to a Resource

NSGs can be associated with either a virtual network subnet or a NIC. The process:

- Go to the subnet or NIC within your resource.
- Locate the **Network Security Group** section in the settings.
- Select the NSG you created and associate it to enforce the rules.

3. Create an Application Security Group (ASG)

An Application Security Group simplifies the management of NSG rules by grouping VMs. Here's how to create one:

- In the Azure portal, search for **Application Security Groups**.
- Click **Create**, choose a resource group, provide a name, and set the region.
- An ASG can be referenced in NSG rules.

4. Associate an ASG to a Network Interface Card (NIC)

To assign an ASG to a VM's NIC:

- Navigate to the VM in Azure and go to the **Networking** settings.
- Choose the appropriate NIC and associate it with the ASG.

5. Create and Configure NSG Rules

NSG rules define how traffic flows to and from a resource. Follow these steps:

- Go to your NSG and access **Inbound** or **Outbound Security Rules**.
- Create a new rule by specifying:
 - **Priority**: Lower numbers have higher precedence.
 - **Source/Destination**: IP, ASG, or any.
 - **Port and Protocol**: Define specific or all ports and protocols.
 - **Action**: Allow or Deny traffic.

6. Implement Virtual Network Flow Logs

Flow logs record information about IP traffic flowing through NSGs. To enable them:

- Open **Network Watcher** in the Azure portal.

- Enable **NSG Flow Logs** for your desired NSG, select a storage account, and configure retention settings.
- These logs are invaluable for auditing and diagnostics.

7. Interpret Virtual Network Flow Logs

Flow logs capture details like source and destination IP addresses, ports, protocols, and rule actions. You can:

- Use Azure Monitor to parse logs for trends or anomalies.
- Employ Traffic Analytics for advanced insights.

8. Interpret NSG Flow Logs

These logs help understand traffic allowed or denied by your NSG:

- Examine fields like **Action** (allow/deny), **Rule** (specific NSG rule), and traffic details.
- Analyze the logs to troubleshoot issues or enhance security configurations.

9. Validate NSG Flow Rules

You can validate NSG flow rules using the **NSG Diagnostics** feature in Network Watcher:

- Test specific IPs, ports, and protocols to ensure that rules are working as expected.
- This helps confirm that only permitted traffic is flowing.

10. Verify IP Flow

Azure's **IP Flow Verify** tool is useful for testing traffic:

- Provide details like source/destination IPs, ports, and protocols.
- The tool verifies if traffic is allowed or blocked based on the NSG rules.

11. Configure an NSG for Remote Server Administration (Including Azure Bastion)

To securely manage servers:

- Configure NSG rules to allow Azure Bastion or RDP/SSH traffic to your VMs.
- Ensure access is limited to specific source IPs for enhanced security.
- For Bastion, no public IPs are required—use the private interface.

12. Implement and Manage Virtual Network Security Using Azure Virtual Network Manager

Azure Virtual Network Manager provides centralized management for network security:

- Define security configurations and deploy them across multiple virtual networks.
- Use features like **Security Admin Rules** for global enforcement of policies.
- Monitor compliance and identify vulnerabilities within your network setup.